

MomentsVR: Accessible Immersive Spaces for Authoring Personal Narration

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Abstract

Migration often fragments cultural heritage, distributing it across geographies and generations. While immersive experiences have the potential to bridge these cultural divides, current immersive tools require technical expertise that excludes many migrants from narrating their own stories. We present MomentsVR, an accessible authoring system that empowers migrants to create and share immersive narratives of their personal experiences. Through importing surround imagery, 2D and 3D content, and ambient audio, users can reconstruct meaningful places and narratives, linking them with teleporters for fluid narrative transitions. The system also supports audio annotations and visual emphasis techniques, such as blurring, focusing and warping to highlight subjective experiences. Our work showcases MomentsVR as a means of democratizing immersive storytelling, allowing migrants to curate, share, and reimagine their cultural experiences across immersive spaces.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing systems and tools.**

Keywords

Migration, Personal Narration, Interaction, Immersion, Virtual Reality

ACM Reference Format:

Nabila Chowdhury, Anna Offenwanger, Theophanis Tsandilas, Fanny Chevalier, and Syed Ishtiaque Ahmed. 2025. MomentsVR: Accessible Immersive Spaces for Authoring Personal Narration. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*,

October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 5 pages.
<https://doi.org/10.1145/3715070.3749270>

1 Introduction

Cultural heritage is actively shaped by migration, transitions between cultures, and the reconstruction of identity across generations [14]. Migration often fragments and disperses cultural practices, prompting migrants to adapt through storytelling, artifact preservation, and rituals within host countries to maintain their cultural identity [5, 7]. Physical and virtual immersive environments such as VR and AR are a powerful medium to consolidate these fragmented stories into interactive narratives [32, 33]. Prior research has explored how immersive storytelling can act as a digital space for fostering understanding across cultural boundaries by placing users in the perspectives of others' culture [8, 22, 30, 33, 37]. This is especially significant for migrant communities, where storytelling serves not only as a means of preserving cultural heritage but also as a bridge to new forms of cultural belonging and continuity in the host country [12, 21]. However, the creation of such immersive environments is often limited to those with advanced technical skills (e.g., programming and 3D modelling), resulting in narrative power being concentrated among those with technological expertise. This dynamic can inadvertently shape whose memories are represented and how migrant stories are told [12].

While co-design initiatives attempt to democratize this process, they are not without challenges, including the risk of homogenizing diverse cultural experiences into standardized forms that may obscure subcultural nuances [30]. For migrant communities whose personal experiences are deeply intertwined with place, identity, and collective storytelling, this gap in authorship restricts their capacity to represent their own histories and transitions [27]. Empowering migrants to directly author immersive experiences could transform how memory and migration are represented, allowing for richer and more nuanced immersive spaces that capture the fluidity and complexity of transnational lives [12]. Our aim is to develop interactions that enable migrant individuals and families to utilize immersive technologies for memory curation and storytelling.

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CSCW Companion '25, Bergen, Norway

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ACM ISBN 979-8-4007-1480-1/25/10

<https://doi.org/10.1145/3715070.3749270>

We propose an immersive narrative authoring system, ‘MomentsVR’, designed to democratize the creation of personal immersive narration. Our system moves towards enabling users to intuitively share stories in immersive environments using everyday artifacts and narration. While prior systems for immersive narration have focused on content creation or generalized VR experiences (e.g., [30, 33]), they often lack accessibility for first-time users, especially those not familiar with immersive technologies. MomentsVR addresses this by enabling novice users to author personal, spatial narratives through accessible workflows that support personal and publicly available media, and by supporting multilevel hardware collaboration between more familiar desktop environments and the VR platforms. This approach empowers migrants to directly craft and share their narration, and fosters agency in how experiences are curated.

2 Background: Migration, Narration & VR

HCI researchers have focused on understanding the technological needs of migrants extensively through information and communication technologies (ICTs) to support functional needs of integrating into the host country, such as building social capital, education, community, etc. [15, 18, 31]. While such functional needs are crucial, viewing migrants’ technological needs from a homogeneous point of view limits understanding nuanced and everyday practices of nurturing cultural values in a host land. Such practices are carried out via curating personal and collective memory, history, artifacts and rituals. [9, 16, 26]. Social science researchers have explored how migrants utilize ubiquitous technologies, such as taking pictures on mobile and sharing them over social media as a form of resilience, holding and sharing cultural identity [17, 29]. We extend this exploration by seeking to understand how migrants can benefit from having an authoring tool that can support memory and culture curation in an immersive environment.

Immersive technologies (e.g., VR) can be reimagined as a digital space that fosters various cultural elements to enrich memory and story narratives for migrants. Instead of focusing on migrant struggles, immersive technologies can highlight cultural richness, aligning with immersive capabilities to enhance storytelling and contexts [24, 25, 32]. Museum and HCI studies show that VR environments can utilize artifacts, photographs, and historical reconstructions to deepen cultural understanding [10, 13, 36]. VR can recreate spaces and offer sensory-rich experiences that go beyond traditional displays, enabling migrants to represent their cultural identities and experience immersive environments curated by others [13, 19, 30, 41]. However, this requires authoring tools that help migrants use daily stories, artifacts, and rituals to build immersive narratives.

Previous research has shown how migrants engage with enriched environments often curated by experts [19, 34, 35], and other studies seek to foster such engagement through participatory and co-design processes [6, 20, 28, 30, 33, 38, 39]. Participatory and co-design are powerful tools to design equitable technology, but often, such design does not go beyond the study duration. This limits our understanding of how migrants might use such technologies in their daily lives. The challenge of creating immersive experiences for non-programmers has been acknowledged by previous research, and

different systems have been created to support novice creators in making content, focusing on aspects such as authoring interactivity for games [11] and instructional videos [23]. Previous work has also explored supporting novice authors in creating immersive AR narratives through the use of household IOT devices [40]. We extend this work to support novice users (migrants) in authoring spaces for curating narration and creating new forms of cultural belonging, specifically by exploring the incorporation of artifacts, personal memory, and personal narrative in the creation of immersive spaces.

3 System Design of ‘MomentsVR’

MomentsVR is an immersive authoring tool that supports users in recreating memories and stories utilizing personal photographs and artifacts, including recorded narration and ambient audio. To make MomentsVR more accessible, we designed a desktop interface which complements the immersive one, enabling users to work on their stories even if they do not have access to immersive technologies. Below, we describe the features of MomentsVR. Figure 1 outlines a step-by-step workflow of how these features support a novice user in creating a narration.

3.1 Narrative creation support

The interactions in MomentsVR support users in creating narrative through three interconnected stages: recreating narratives, enriching the narratives, and transitioning between different parts of the narratives.

Recreating Narratives. To create narratives in the MomentsVR, users must be able to recreate scenes from their own lives and guide through transitions between them in the system. Our application breaks a narrative into a series of “moments,” with each moment grounded in a surround image (Figure 1a) wrapped onto a sphere mesh. While this works best with equirectangular projected 360° images, ordinary images can also be used. Environmental sounds can be recreated by importing ambient audio. Authors can also incorporate 2D images (Figure 1e) and 3D assets to enrich the context of the environment. A challenge in authoring is that migration can cause the loss of artifacts, such as photographs being left behind in the home country. We, therefore, experiment with creating experiences with publicly available artifacts. Users can import artifacts on the internet and surround imagery from Google Maps.

Enriching Narratives. Users can express their own personal experiences and memories through the use of annotation and emphasis. As 360° surround images can easily over-include information, the author chooses what parts of the surround image to focus on through the use of focusing and blurring parts of the image (Figure 1d). They can further emphasize parts of the surrounding environment by flattening and extruding the related sections of the sphere mesh to make them more prominent (Figure 1c). This requires users to tell stories and describe their experiences through the use of audio recording. Audio recordings exist within the environment as a generic sphere mesh or can be attached to specific 2D or 3D assets which the author wishes to annotate (Figure 1f). Attached audio annotations play when the viewer interacts with the object.

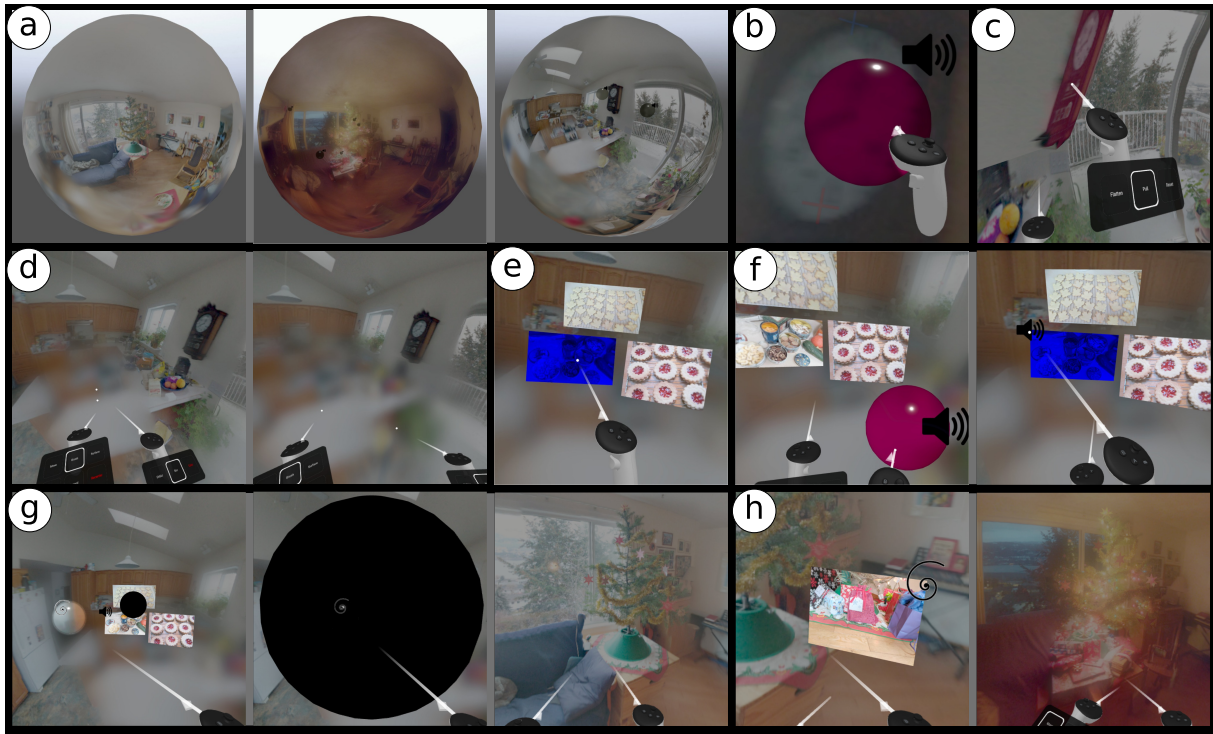


Figure 1: Moments supports authors in telling personal stories in an immersive environment. (a) The author creates a story of their experience of Christmas in a series of ‘moments’, using 160 images to recreate their living room (day and night), and kitchen. (b) They finish the immersive scene by adding the ambient audio of the ticking clock (marked by a generic sphere and audio symbol). (c) Because it is important to their experience, the author emphasizes the kitchen clock through extruding the mesh to make it more prominent, and (d) uses blurring to de-emphasize the unimportant kitchen clutter. To add details, the author (e) arranges a series of images and (f) records and attaches audio annotations. (g) The author then adds teleporter to enable transition from the kitchen to the living room, and (h) from there to Christmas eve. They contextualize the last transition by attaching it to an image of presents.

Transitioning within Narratives. One of the most valuable aspects of immersive environments is their ability to “teleport” a viewer to another place and time. To transition between different environments, users can add “teleporters”. Teleporters are objects within the environment that, when dragged, generate a teleport drop target (Figure 1g) which transitions the user to a new scene. Teleporters can either exist as a generic sphere or be contextualized by attaching them to 2D images (Figure 1h) or 3D models.

3.2 User Scenario and Experiences

As a preliminary evaluation of the interactions developed in *MomentsVR*, the first and second authors used the system to author four narratives, two of which we discuss here.

German Canadian Christmas. The first narrative (Figure 2) makes extensive use of personal archives and imagery. The equirectangular surround images were created using an Android Mobile phone and the Google Camera App. Blurring proved to be extremely effective not only as a means of bringing subjectively important content into focus (clock, Christmas tree) but also in concealing seams in the image as well as unwanted content (e.g. feet). The snow scene depicted in Figure 2c shows a set of photos arranged along

a temporal axis and provides rich data on the progression of time from a single position. The author found it personally fascinating to reflect on the passage of time as she arranged these images.

Pohela Boishakh in Toronto and Bangladesh. This narration (Figure 3), experiments with authoring personal stories from memory in the absence of personal media. The author acquired key scenes from Google Maps street view using a free download tool [1]. In Figure 3a, the images used serve generically to show the food and decorations that would be encountered at the time of the celebration. The author experienced such celebrations during childhood in their homeland but did not have access to photographs with which to create this narration. In the scenes, the augmenting images are annotated to explain their significance.

3.3 Implementation

We implemented the MomentsVR interface in Javascript using Three.js [2] and the WebXR API [4]. This enables the application to run on a variety of devices as the WebXR API is increasingly supported across browsers and headsets. We focus our implementation and testing on the Chrome browser and the Meta Quest

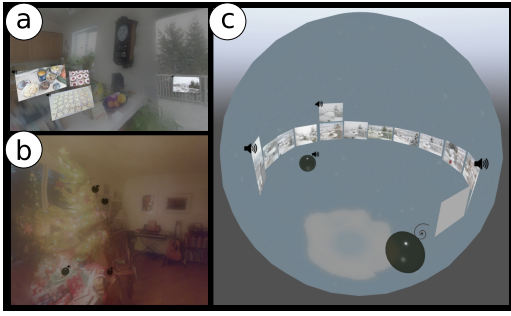


Figure 2: German Canadian Christmas: This immersive narrative describes Christmas from a German-Canadian perspective. (a) The story starts in the kitchen, where the author describes some of their experiences and what made their Christmas different from their peers. From the kitchen, the image on the right would teleport the viewer to (c) a snowy scene with an annotated collection of photos from the balcony, and the author reflects on snow, winter, and Christmas over many years. The portal at the end of the gallery will bring the viewer back to the kitchen, where they can transition to (b) the living room, where the author describes Christmas Eve and reminisces about childhood memories.



Figure 3: Pohela Boishakh in Toronto and Bangladesh: Narrative of the Pohela Boishakh celebration as experienced in Toronto compared with Bangladesh. (a) The narrative starts on a street in a large Canadian city which Bangladeshi people decorate for the celebration. 2D images show the decorations and food that are found at the festival. (b) Viewers transition to Bangladesh, where the moment captures how the people prepare the roads and buildings with designs for the celebration. (c) In the early morning of Pohela Boishakh, people gather with big decorations for a rally. To set the tone of the festivities, this moment includes music from the festival.

headsets, which are comparatively common and affordable standalone devices. Narrations in MomentsVR can be edited either from the desktop interface or the immersive VR interface. Due to file API restrictions, functionalities for uploading files and managing projects are restricted to the desktop interface. Story files are stored locally on the author's computers, ensuring they can maintain control of their artifacts.

The backend of the MomentsVR system is implemented in Node.js, and uses the WebSockets API [3] to enable multiple users and devices (i.e. a computer and a standalone headset, or two users using headsets) to synchronously collaborate on a story.

3.4 Preliminary Feedback

We evaluated MomentsVR with an experienced UX designer who also has a migration background. The evaluation consisted of one hour reviewing examples created by the author and an additional hour collaboratively creating a narrative. As a first-time user, the participant found the system intuitive once they began curating their own narration and grasped the workflow. They provided the following feedback for the next iteration:

- The participant found the system easy to use and surprisingly effective in evoking a sense of place, even with basic surround imagery. The participant found “when I look at those images on Google it just images. But when in the 3D world that I created, I had that type of feeling that I feel like I can touch it, [...] I could feel it.”
- They suggested that surround audio would enhance immersion; while this feature exists, it was clearly not discoverable. This indicates a need for further exploration of immersive audio editing interfaces.

4 Future Work & Evaluation

MomentsVR serves as both a demonstration of a potential toolset and a design probe to elicit from our target audience what a complete toolset might look like. Through future evaluation, we aim to identify a concise toolset that supports a wide range of compelling experiences while remaining simple enough to use without extensive training.

For next steps in this project, we will extend our evaluation with additional participants. Participants will be asked to view example stories and try the authoring system. This process will serve as a basis to understand how our target users envision MomentsVR supporting their narratives and what additional features they might need to express their stories. As future work, we also aim to explore collaborative storytelling in family and community settings. The collected feedback will guide future iterations of the toolset to enhance support for culturally rich narratives and promote accessible technology design for marginalized communities.

Supplementary Materials

The Moments authoring system and gallery are available online at <https://offenwanger.github.io/moments-demo/>.

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